Research Methodology and AI Disclosure

CIF Ancient Systems Test Series
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Research Approach

The CIF Ancient Systems Test Series uses historical and archaeological case studies to stress-test the Coherent Inheritance Framework (CIF) — a candidate cross-domain framework for analyzing how information, knowledge, and capability are transmitted across discontinuities in time, technology, and culture.

Each case study follows a disciplined methodology:

1. **Evidence Classification** — All claims are tagged as ESTABLISHED (consensus scholarly position), PLAUSIBLE (defensible interpretation with some scholarly support), or SPECULATIVE (hypothesis or extrapolation requiring further validation).

2. **Source Grounding** — Every factual claim is traced to a specific published source (peer-reviewed papers, monographs, archaeological reports, or authoritative historical texts). Full bibliographies are provided in `sources.md` files for each case.

3. **Falsification Checks** — Each case study ends with a mandatory falsification check that identifies what evidence would disprove or significantly challenge the CIF analysis presented.

4. **Framework Consistency** — All case studies use the same analytical structure: Five Inheritance Layers (data, structure, method, institutional, living), six CIF axioms, and a shared verdict taxonomy.

5. **Counterexample Inclusion** — Cases that challenge or refine CIF (e.g., Greek Fire's permanent loss, the Antikythera Mechanism's institutional collapse) are included deliberately, not excluded.

6. **Transparency and Correction** — Version history is preserved via GitHub. Errors discovered after publication are documented in the CHANGELOG and corrected in subsequent releases with clear notes about what changed and why.

AI-Assisted Research and Drafting

AI-assisted research and drafting tools were used during the development of this project.

This includes:

- Literature search and source discovery
- Outline generation and case study structuring
- Drafting prose for case studies, synthesis documents, and framework explanations
- Citation formatting and bibliography management

- Code generation for repository automation (PDF conversion, metadata files)
- Logical consistency checking and counterargument identification
- Summarization and cross-case pattern analysis

All source selection, evidentiary classifications, framework interpretations, revisions, and final publication decisions remain the responsibility of Derek Hone and Remnant Fieldworks Inc.

Governing Principles for AI Use

AI tools are treated as **research assistants**, not as authorities or independent validators.

The following principles govern AI use in this project:

1. Evidence Discipline Over AI Output

AI can suggest sources, draft prose, or propose interpretations — but **evidence always governs the final conclusion**. If an AI-generated statement cannot be traced to a real, verifiable source, it is not published.

2. Verification of Citations

AI can invent citations, confuse dates, or blend competing theories. **All citations are verified against actual published sources** before inclusion. No unverified AI-generated references are used.

3. Disclosure of Uncertainty

AI tends to produce confident-sounding prose even when evidence is weak. **Speculative claims are explicitly tagged as SPECULATIVE**, and uncertainty is preserved rather than papered over.

4. Human Responsibility

AI is not listed as a co-author, reviewer, or independent validator. **Derek Hone remains fully responsible** for the accuracy, integrity, and scholarly quality of all published work.

5. Counterexample Inclusion

AI can exhibit confirmation bias by emphasizing supportive evidence and downplaying contradictions. **Cases that challenge CIF are deliberately included** (e.g., Greek Fire, the Antikythera Mechanism), and AI-suggested exclusions of inconvenient evidence are rejected.

6. Public Correction

When errors are discovered — whether introduced by AI, human oversight, or misinterpretation — they are **documented publicly in the CHANGELOG** and corrected in subsequent releases.

What AI Cannot Do

AI tools cannot:

- Conduct original archaeological fieldwork or archival research
- Provide independent peer review or scholarly validation
- Establish that CIF is proven science or consensus theory

- Replace the judgment required to classify evidence as ESTABLISHED vs. PLAUSIBLE vs. SPECULATIVE
- Determine whether a case study should be included or excluded based on its fit with CIF

These remain human responsibilities.

Credibility Standard

The legitimacy of this work does not rest on **whether AI was used**.

It rests on whether the work is:

✅ **Transparent** — Sources, reasoning, and version history are public and inspectable
✅ **Source-grounded** — All factual claims trace to real published scholarship
✅ **Falsifiable** — Each case includes conditions under which the analysis would fail
✅ **Accountable** — Errors are corrected publicly, not hidden
✅ **Honest about limitations** — This is an independent research program, not peer-reviewed consensus science

AI is a tool that can accelerate research, drafting, and organization — but it does not replace the discipline required to meet these standards.

Peer Review and Validation Status

This work has not undergone formal peer review by an academic journal or institution.

It is an independent research program conducted by Remnant Fieldworks Inc. and published openly via GitHub and Zenodo for public inspection, critique, and use.

Readers, scholars, journalists, and other researchers are invited to:

- Inspect the sources and reasoning
- Challenge the evidentiary classifications
- Propose alternative interpretations
- Identify errors or omissions
- Test CIF against additional case studies

GitHub issues and pull requests are welcome for substantive critiques or corrections.

Publication and Preservation

GitHub serves as the **living development record**, showing how the project evolves over time. It allows readers to inspect version history, compare releases, and see corrections.

Zenodo serves as the **formal archival record**, assigning permanent DOIs to specific released versions and preserving them for long-term citation and access.

Together, they ensure that:

- The work is publicly accessible and citable
- Version history is preserved
- Corrections and refinements are documented
- Future readers can verify claims against original sources

This aligns with CIF's own principles: **preservation requires transparency, redundancy, and active stewardship**, not passive storage or hidden processes.

Final Statement

We use modern tools aggressively, but we do not outsource truth.

AI is a legitimate and powerful aid for research, drafting, organization, and analysis — but the final responsibility for accuracy, honesty, and scholarly integrity rests with the author.

The strength of this work is not that it was created without AI.

The strength is that it is **transparent, source-grounded, falsifiable, and accountable** — regardless of the tools used to produce it.

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